



The role of the chatbot on customer purchase intention: towards digital relational sales

Letizia Lo Presti¹ · Giulio Maggiore¹ · Vittoria Marino²

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Abstract

The purpose of this paper is to explore the extent to which conversation with a chatbot on an official website can change the value perception of products and influence the customer purchase intention. The study also investigates the role of chatbots on brand familiarity and the influence of previous experience of interaction with chatbots on customer buying intention. Two laboratory experiments were conducted to evaluate the impact of chatbot use on customer purchase intention. Chatbots on websites shift the value perception of hedonic and utilitarian goods towards respectively utilitarian and hedonic values. Chatbots contribute to the brand awareness reducing the effect of familiarity on the customer purchase intention. Retailers should include effective chatbots while operating and designing shopping websites to stimulate the shopping intention of consumers. Companies should design and carefully manage chatbots by monitoring user involvement and quality of customer experience in order to feed a calibrated continuous improvement process on company objectives. The paper represents a first step approach in addressing e-service agents toward the online customer shopping experience. Digital assistants may change the online shopping context increasing the hedonic value experience but at the same time reducing the time for decision making by reinforcing the utilitarian value perception of the utilitarian and hedonic products.

Keywords Chatbot · Conversational commerce · Customer purchase intention · Value perception · Brand familiarity

✉ Letizia Lo Presti
letizia.lopresti@unitelmasapienza.it
Giulio Maggiore
giulio.maggiore@unitelmasapienza.it
Vittoria Marino
vmarino@unisannio.it

¹ Department of Law and Economics, University of Rome Unitelma Sapienza, Rome, Italy

² Department of Law, Economics, Management and Quantitative Methods, University of Sannio, Benevento, Italy

1 Introduction

The growing presence of chatbots on the official websites of the brands demonstrates the importance that is given to contact with the customer. Accenture reveals that 56% of companies consider conversational bots a disruptive technology that can potentially deliver a secure return on investment with a minimum of effort (Accenture, 2018). They can be defined as “*system-based autonomous and adaptable interfaces that interact, communicate and deliver service to an organization’s customers*” (Wirtz et al., 2018, 909). Chatbots, like all service robots, can elicit conversations for commercial purposes. Chatbots have certain characteristics that differentiate them from employees: they act as part of systems, they update continuously, they use learning systems and have a practically infinite memory. It takes a chatbot only a moment to do what requires continuous updating for employees and the final results still depend on their individual backgrounds and learning abilities (Wirtz et al., 2018).

With new technologies such as A.I. applications and chatbots for commercial purposes, the possibilities offered through interaction with the brand/organization have been expanded and innovated, making information sorting more controllable, faster and more customizable. Conversational marketing is the new branch of marketing that has conversation as its object, used as a lever that can motivate the customer to take an interest in the product and encourage purchasing. In this regard, conversational commerce develops, in the retail sphere, an extension of social commerce, that puts the conversation between customer-brand/organization at the centre of the relationship (Brusch & Rappel, 2019), and follows the entire customer journey from the acquisition of information to post-sales via a conversational agent (chatbot) or a live operator who can interact at any time with the customer (Hoyer et al., 2020). The heart of conversational commerce is the role of conversation. Man–machine conversations or conversations through live operators take on strategic importance for lead marketing. Ciechanowski et al. (2019) for example have studied human–machine interaction to detect the effectiveness of user responses to the different interface forms with which they interact. If the offline conversation with the seller (salesperson) has always been appreciated by the consumer for its functional (suggestions, time savings, better purchasing decision) and social benefits (Reynolds & Beatty, 1999), in the digital environment, conversation in the virtual space creates an experiential microenvironment in which the values of hedonicity and functionality can act as a stimulus for the purchasing process (Chiu et al., 2014; Richins, 2012; To et al., 2007). Indeed, as affirmed by Mimoun et al. (2017), beside and beyond the mere presence of such an agent, the virtual interaction is a source of perceived benefits for Internet users on commercial websites that improves e-consumer productivity in the shopping experience. In the face of this strategic role and its complexity, it is important to understand the implications of the presence and use of chatbots on the official brand website. Chatbots have been extensively analysed in the marketing literature, but many studies move towards understanding the importance of chatbots or virtual assistants in the field of customer care (Sangle-Ferriere & Voyer, 2019) or

for intentions to purchase but without including the context in which the interaction takes place and without detecting its importance for the perception of value of the product (Chung et al., 2020). In fact, there have been limited attempts to study the effect that conversation through chatbots or live chats has on the buying process during online shopping on the website (Kim et al., 2012; Song & Christen, 2019; To et al., 2007).

Given the literature to date, the purpose of this study is to shed light on this aspect by better understanding how interaction with a chatbot works on the website level and how it encourages the intention to purchase by altering the perception of the value of the product/service. While previous research was confined to exploring the intention to purchase according to the type of interaction with new technologies (for example for exclusively hedonic services) (Danckwerts et al., 2019), it has been conceptually and empirically documented that the intention to purchase can be elicited through interactive activities and the ability to know how to give useful information to the potential buyer, and that the perceived value of a product depends on the context in which one comes to experience it (Mathwick et al., 2002). Furthermore, chatbots have been analysed with reference to hedonic products (Danckwerts et al., 2019) without ever considering their importance for functional products. The research perspectives identified in the literature raise the question of the importance of the use of chatbots for the intention to purchase and for the perceived value. To address this issue, this study investigates the role of chatbots in promoting the intention to purchase by becoming auxiliary channels in an omni-channel approach. In addition, the study verifies whether interaction with a chatbot alters the perception of the value of the product after its use. Specifically, the effectiveness of the chatbot on purchase intentions and on the perception of value is analysed with reference to the hedonic and utilitarian value of the product.

The proposed theoretical framework offers insights for both scholars and practitioners. From a theoretical point of view, this research echoes the research of To et al. (2007) who hoped for a greater understanding of perceived value during the online shopping experience. In addition, it addresses a greater understanding of the role of conversation concerning the perceived value of the product during the customer contact process (Sampson & Chase, 2020). So this study responds to a current line of research on the role of conversation in relational marketing strategies and extends the understanding of the role of chatbots in marketing services with some insights coming from the results of this research.

Moreover, knowing the effect of the conversation on the purchasing process and on the change in the perception of the utilitarian and hedonic value of the product could be useful to those who are thinking of investing in these channels; above all it would allow them to understand how to use these channels in accordance with the marketing objectives that companies want to reach and the reactions they want to receive from the customer. Can chatbot conversation tolerate and support new levels of customer interaction with products and services by changing the perception of the product and stimulating the intention to purchase? By addressing this question, this study aims to shed light on chatbots during online shopping and their role on the perception of value of the product/service during the online experience.

2 Literature review

2.1 Conversational commerce: the value of digital-based relational sales

In the field of digital commerce, a succession of transformations has seen the emergence of new shopping models that have changed online purchasing methods and returned greater customer satisfaction. In this sense, Leitner and Grechenig (2008) call "social shopping" and "collaborative shopping network" the use of social networks during the phases of purchase. These models are based on the use of social media as privileged channels not only for the exchange of information but also today for the purchasing of products. For example, the use of Instagram by fashion companies is a clear example of the "online network market" (Stephen & Toubia, 2009). This possibility of integrating the relational space with the purely commercial one has allowed us to restore an added value that has not been fully explored to date. The exchange of information (through reviews, comments, likes, shares) used during the customer journey has concretized the occurrence of the so-called s-commerce: the use of social media for the purchase (Yeon et al., 2019). With A.I. applications to chatbots for commercial purposes, the possibilities offered through interaction with the brand/organization have been expanded and innovated, making information sorting more controllable, faster and more customizable. Conversational commerce consists of an interaction process aimed at involving and selling products or services based on conversations mediated through digital messaging systems.

Different definitions of conversational commerce have followed one another since 2016 but without reaching a univocal definition. In this sense, great importance is given to conversation "as an interface for commerce". It transforms the way of interacting and communicating by reducing the distances between customer-company. Since it is based on conversation, it makes the dialogue with the brand/organization familiar and confidential, lowering the physical and psychological barriers that sometimes prevent the customer from responding adequately to requests and hinder his involvement.

Despite the abundant literature on social commerce (e.g. Leitner & Grechenig, 2008; Stephen & Toubia, 2009), to date little has actually been investigated regarding the potential of instant messaging, especially with regard to the effect this can have on the perceived value of products/services and the intention to purchase during online shopping. Chung et al. (2020) for example have studied the consumer's perception during the use of the chatbot with luxury brands and have discovered that chatbot services offer interactive and engaging experiences between brands/customers that facilitate the perception of satisfaction despite the fact that the quality of the communication has not yet been fully verified. Despite this, little has been studied so far in the context of conversational commerce.

Yet in many countries these applications have been enhanced to become real touch points throughout the customer journey (Lemon & Verhoef, 2016) not only for lead generation activities but also for sales. In particular, these applications are able to influence what in literature is referred to as *the first moment of truth*,

namely the moment in which the product/brand is able to actually attract attention and arouse the specific interest of the consumer, thus influencing the purchase.

Studies were begun to understand the use of instant messaging as external channels, for example, during the offline shopping of a customer who shows a particular attachment to the chat thus modifying, developing and re-semanticizing the real concept of “shopping” (Pantano & Gandini, 2017). Pantano and Gandini (2017) speak of a “meta-private” phenomenon, which is an individual approach but orientated towards social consumption. One of the advantages found in the use of instant messaging in business is that of being an “essential selling tool”. Chopra (2019) investigated the motivations to use AI tools such as chatbots, augmented reality and voice assistants by younger Indian shoppers. Ease of use, competency in performance and satisfaction, trust and a rewarding experience are considered the main motivations to use AI tools.

Diversely, Chung et al. (2020) studied the use of chatbots in the luxury fashion retail brands and revealed that chatbots have a hedonic component (interactivity, entertainment, trendiness) but also a utilitarian component (customization, and problem-solving components) thanks to which performance can be measured. Moreover, the authors demonstrated that chatbot e-service provides interactive and engaging brand/customer service encounters. Despite this, little has been studied to date about the effect this may have on the intention to purchase. Today, in fact, success as a customer-friendly brand at any stage of the customer journey offers great opportunities, especially as it can support/strengthen all the touch points and in particular those points that could help the product to be part of the so-called customer consideration set, and actively help him evaluate the feasibility of a possible purchase.

Moreover, in a logic increasingly orientated to intensifying the customer experience—who today wants to know more and see more about the brand, and actively participate in the brand’s initiatives before and after the purchase—the chatbots become a channel capable of expanding the consumer experience by making the brand available to a chat. Fournier (1998) had already advocated a quality brand relationship with the customer based on intimacy and commitment. It does not represent the final stage of the purchasing process but it becomes the means by which to proceed to the purchase. In the light of these considerations, therefore, knowing how to appreciate the value of conversation during a potential purchase process, can help to understand the value that organizations should give to conversations in their marketing strategies and the resources that should be used to make the tool effective to the strategic orientation of the organization and how to finalize it to the acquisition or maintenance of the relationship (Iankova et al., 2019).

2.2 Digital context and value perception of hedonic/utilitarian products

The scope of the perceived value of a product has always been the subject of intense debate since it determines the marketing mix strategies that will be implemented by marketing managers. The value is perceived by consumers differently as the variables subject to their evaluation are different. Quality, price, intrinsic characteristics of the product and type of services, return a different interpretation of the value of

the product according to the needs it satisfies. Products can be perceived as more or less hedonic or utilitarian based on the sensations that the product arouses in the consumer's mind. Batra and Ahtola (1991) introduce a two-dimensional (hedonic/utilitarian) approach to understanding consumer behaviour. These two dimensions characterize the perceived value of the product (e.g. Voss et al., 2003; Thamizhvanan & Xavier, 2013; Dhar & Wertenbroch, 2000). In this context, the literature is unambiguous in defining utilitarian products (*utilitarian goods*) as those: “*whose consumption is more cognitively driven, instrumental, and goal oriented and accomplishes a functional or practical task*” (Strahilevitz & Myers, 1998), while hedonic products (*hedonic goods*) are defined as those: “*whose consumption is primarily characterized by an affective and sensory experience of aesthetic or sensual pleasure, fantasy, and fun*” (Hirschman & Holbrook, 1982).

However, according to several authors (Cassia et al., 2015; Gummerus & Pihlström, 2011) the utilitarian and hedonic value of the product is subject to change since the value of a product is always in progress and co-created by consumers during the purchasing process and during consumption. According to, for example, Cassia et al. (2015) and Gummerus and Pihlström (2011), the service-dominant logic approach is the most suitable to express this concept. In fact, according to Cassia et al. (2015) “this approach states that value is not co-created with the customer during consumption (value-in-use)” (p. 4). In this regard, Gummerus and Pihlström (2011), study the concept of value-in-use in the mobile context where hedonic and utilitarian components are mixed together during interaction with a product. For the authors “in use-value” (consisting of emotional, esteem, monetary, convenience, and performance value) stems from a person's experience of using a specific service (p. 526).

Therefore, despite the value proposition of a product, the perception of value during online shopping can change at any point just like the online shopping experience can (Gupta & Kim, 2010). During online shopping the value-in-use is subject to the context in which the evaluation is made (Kohler, 2011; Rose et al., 2012). Other authors have traced the influence on perceived value to the importance of the digital context during the perception of the value of the product. To et al. (2007) showed that online shopping is linked to utilitarian values and still very little to hedonic values (To et al., 2007). This means that regardless of the value that the products have in themselves, the digital environment has components linked to the utilitarian dimension, in turn linked to the acquisition of information on the product-service that supports the utilitarian motivation to purchase the product. Despite this, digital shopping also has an experiential component, linked to entertainment and pleasure (Richins, 2012) which can increase the hedonic value of products, for example, of a purely utilitarian nature, guaranteeing the re-purchase of the product. In this sense, some authors (Moon et al., 2017; To et al., 2007) hope for a greater commitment by companies towards online content that will elicit the emotional dimension of customer involvement.

Thanks to technological innovation and new forms of digital socialization, such as the integration of instant messaging also on online shopping websites, it is possible to provide interactions and sensory stimuli typical of hedonic shopping value (Babin et al., 1994). Chung et al. (2020) demonstrate that chatbot e-services offer

interaction and allow greater involvement between brand and consumer. Other authors (Gan and Wan 2017; Moon et al., 2017) found that key online shopping website attributes influence the cognitive and affective attitudes. Moon et al. (2017) discovered that "Cognitive and affective attitudes are significant predictors of the consumers' online purchase intentions". Chatbots, thanks to their function of providing information, also perform a utilitarian function by answering the questions posed by their users. We can therefore hypothesize that chatbots inserted directly into websites stimulate hedonic values and utilitarian values in both hedonic and utilitarian products during the online purchase process. To et al., (2007, 776) also came to this deduction which leaves the confirmation of this hypothesis to future research developments. We can therefore hypothesize that:

H1a: *the chatbot increases the utilitarian value of hedonic products*

H1b: *the chatbot increases the utilitarian value of utilitarian products*

H1c: *the chatbot increases the hedonic value of hedonic products*

H1d: *the chatbot increases the hedonic value of utilitarian products*

2.3 The role of the chatbot on customer purchase intention

Customer purchase intention is a core issue in marketing management, as it influences the customer's final decision. McLean and Osei-Frimpong (2019), for example, investigated the chat live and showed that the variables that affect such use depend on the context for initiating the chat conversation, namely for supporting search/navigation or supporting decision making. The main goal is to directly make the client not only chat and use the services, but eventually create a product or service purchase, enhancing his/her customer journey. Already Bacik et al. (2017) affirmed the importance of developing several marketing innovations by means of the use of AI. In particular, the authors found that artificial intelligence could be used to trigger a chatbot or live chat communication in order to assure the user as regards his purchase intention. Chung et al. (2020) studied the effect of chatbot services in the luxury retail context on customer satisfaction and demonstrated that digital service assistance tools can help build positive customer relationships even though e-service agents do not fully communicate with customers. For this reason, chatbots can be considered a substitute for customer–service agent communication that can finalize the selling by means of a correct communication based on the five quality dimensions—interaction, entertainment, trendiness, customization, and problem-solving (Chung et al., 2020).

Sands et al. (2020) also come to similar conclusions. The authors investigate service scripts and their role in digital service delivery. They demonstrate that interactions with chatbots if personalized (using specific narrative cues based on entertainment), may optimize emotion and rapport, and therefore customer purchase intent and satisfaction. Insider-Trends (2017) shows that service agents still influence 87% of in-store purchase decisions, and that familiar salespersons are crucial during the purchase (Insider-Trends, 2017). It can therefore be expected that customers chatting with a conversational agent can perceive benefits from the chatbot conversation when browsing on the official website. In the sense of chatbot commerce, the

findings confirm the important positive effect of perceived pleasure and perceived social presence on consumers' purchasing intentions (Han, 2021). The primary motivator for using and purchasing from online shopping sites has been identified as enjoyment (Han, 2021). Chopra (2019) found that ease of use and competency in chatbots influence consumer satisfaction, something that literature affirms to be a strong predictor to customer purchase behaviour (Chung et al., 2020). More formally, the following hypothesis is stated:

H2a: The use of a chatbot while browsing a website has a positive influence on the propensity to purchase in hedonic products

H2b: The use of a chatbot while browsing a website has a positive influence on the propensity to purchase in utilitarian products

2.4 The role of familiarity and previous experience with a chatbot on customer purchase intention

Park and Stoel (2005) found that previous experience with online shopping positively influences consumer attitude during the online buying process. Prior online purchase experience has been found to have a significant effect on purchase intention in the Western context (Shim & Drake, 1990; Shim et al., 2001). Online shoppers who have bought products online are more open and inclined to shop online than others (Lee & Tan, 2003). Shim et al. (2001) found that past satisfactory online purchasing will lead to future online purchasing. So, we can say that a previous experience with a chatbot has a positive influence on the intention to buy a product.

H3a: The previous experience with a chatbot has a positive influence on the intention to purchase.

Familiarity is demonstrated to be a predictor of the customer purchase intention (e.g. Grewal et al., 1998; Laroche et al., 1996). Laroche et al. (1996) demonstrated that familiarity with a brand influences a consumer's confidence toward the brand, which in turn affects his/her intention to buy the same brand. Mimoun et al. (2017) failed to prove that product familiarity positively influences the amount of time spent in navigation and achieving better decision quality and better usefulness were independent of using an animated conversational agent (ACA). Mimoun et al.'s result suggests that conversational agents may have a role in socializing with the brand which lowers the importance of product familiarity.

Araujo (2018) investigates the effect that anthropomorphism design cues and social presence create on overall attitudes, emotional connection and the satisfaction that customers feel with a company by comparing the conversation with a human-like agent and a machine-like agent. The author demonstrated that chatbots influence how consumers perceive the company itself. The conversation with a firm might stimulate the intention to buy accelerating the brand awareness process in presence of unfamiliar brand. In this sense it can be expected that consumers perceive the brand as more familiar and closer when they interact with a conversational agent. Araujo (2018) indeed hoped for future research to investigate how factors such as task complexity and familiarity with the company may

also influence satisfaction with the service provided, and attitudes towards the company. Moreover, Liebrecht et al. (2020) in a study about the informal communication style of a chatbot, found that participants' familiarity with the brand does not influence the customer-brand relation in the chatbot conversation. Since there is no evidence of the moderating effect of brand familiarity in human-to-chatbot service (Liebrecht et al., 2020), chatbots do not assume a different role as communication partner.

Hence:

H3b: familiarity with a hedonic product does not change the purchase intention during the consumer-chatbot conversation.

H3c: familiarity with a utilitarian product does not change the purchase intention during the consumer-chatbot conversation.

We argue that conversation influences value perception of hedonic and utilitarian products and leads to customer purchase intention. We postulate that product familiarity has a different role on the customer's buying intention, i.e. chatbot users are less affected by familiarity products than website users. Finally, previous experience with chatbots does not have any impact on customer purchase intention. To verify our hypothesizes two studies were conducted in the laboratory involving two brands. Study 1 was conducted on hedonic goods (a cruise travel brand was used), while study 2 was conducted on utilitarian goods (a telephone service brand was used).

3 Method

3.1 Subjects

Participants ($N=184$) in this study were young attending college. According to some authors (Lee & Johnson, 2002; Park & Stoel, 2005), college students are especially likely to be potential internet shoppers based on internet shopper demographics. Earlier studies affirm that people who are younger are more likely to have used the internet and have greater purchase intentions for internet shopping (Bailey et al., 2016). Student participation was voluntary. Table 1 illustrates the demographic profiles.

Since we want to investigate the role chatbot conversation on the perception of the hedonic/utilitarian value of hedonic and utilitarian products, two products canonically considered hedonic and utilitarian, the brand of a cruise travel—that sells cruises and vacations trip for tourists—and a telephone service brand—that sells telephone services—were chosen for the experiments. Since the value in use of the product is a function of the context of online shopping (Liao et al., 2016; Turel et al., 2010) we noted the degree of hardness and functionality of the product as perceived by the potential consumer before and after the start of the experiment. 104 participants were involved in the hedonic goods experiment (study 1); 80 participants were involved in the utilitarian goods experiment (study 2) (see Table 1).

Table 1 Demographic profile

Demographics	Study 1 experiment on hedonic product		Study 2 experiment on utilitarian product	
	Frequency	Percent	Frequency	Percent
<i>Sex</i>				
Female	54	51.9	40	50
Male	50	48.1	40	50
<i>Age</i>				
20–21	92	88.46	70	87.5
22–23	12	11.54	10	12.5
Total	104	100	80	100

3.2 Design experiment and overview of the procedure

A mono-factorial experimental design using two conditions of online interaction (website and chatbot) was employed in the experiment. The condition of website is used as the control condition (baseline). In the chat condition individuals were invited to chat about hedonic/utilitarian products. This request was absent in the control condition.

The study was run in an academic computer laboratory since November to December 2019. Participants in the experiment first answered a set of questions about brand familiarity and their preceding experience with a chatbot that were then used to assess previous experience with chatbot and the rate of brand familiarity. In the questionnaire containing the instructions, two questions, “what is your level of familiarity with the *x/y* brand?” evaluated on the 5-point Likert scale (1 = not at all, 5 = very strong) (as in Park & Stoel, 2005) and “Have you ever used a chatbot?” (1 = yes or 0 = no) were used to assess previous experience of chatting by means of chatbots and to evaluate the rate of brand familiarity for the assigned utilitarian/hedonic brand. Participants were also asked to evaluate their perception regarding the value of products/services of the brand object of the experiment before starting the experiment using the 5-point Likert scale (1 = strongly disagree, 5 = strongly agree). Then participants were instructed to interact with a conversational agent and to browse the official web site (the control group could only navigate on the official website). At the end of the experiment, participants in both conditions were asked to fill out a short survey featuring socio-demographics and to assess customer purchase intention (Kim & Lennon, 2000). Furthermore, participants were asked to evaluate once again their perception regarding the value of products/services of the brand after using the chatbot or navigating the website. To evaluate the contribution of chatbots to purchase intention two control-questions are asked: “How likely is it that you will buy what you found today on the internet in the next 12 months” and “How likely is it that you will buy the product/mobile phone service seen on the telephone service brand website next year”. A 5 point Likert-type scale was used to measure the accordance of respondents. Instructions indicated that the participant was to

interact with the conversational agent and/or navigate on the website for no more than 6 min.

3.3 Stimulus

Conversation. Before starting the interaction with the chatbots, the participants read a set of instructions indicating the task they needed to complete. In these instructions, participants exposed to the conversational agent condition were told that they had to interact with a conversational agent. Participants exposed only to the website were told they had to browse the official websites.

The hedonic and utilitarian value of the products were evaluated in the experiment both before the stimulus and after the stimulus (use of chatbot) in both the hedonic and utilitarian product experiments. Participants were invited to the computer lab and randomly assigned to one of the two types of brand. The researcher explained the procedure. Then, participants were first asked to rate brand familiarity (as in Park & Stoel, 2005) for the assigned two types of brand (cruise travel and telephone service). After completing these items, on the next page, a scenario was given to participants asking them to browse the assigned web site with the intention of purchasing and chatting with a conversational agent (the latter only for the experimental group). Each group of participants was assigned to browse the official web site to search for a brand product/service item for 6 min. The assigned browsing time was estimated in a pre-test and was based on the investigator's observations. After searching for the product/service on the assigned website, participants answered the survey questions.

4 Results

4.1 Chatbot and value product/service

Table 2 illustrates the summary of customer shopping experiences across the two conditions. It indicates that the perception of utility after the use of the chatbot is different both in the 1st study and in the 2nd study compared to that obtained after just browsing the website. Independent sample t-tests were conducted to compare the two groups of respondents (experimental and control group). From this comparison, it was possible to determine whether the two groups are significantly different in their product value perception, familiarity and intention to purchase.

For detecting the reliability of the scale in measuring the customer purchase intention a reliability analysis was conducted on the survey responses. The Cronbach's alpha coefficient, is satisfactory ($\alpha=0.70$). The chatbot confers a greater perception of functionality both to the hedonic product and to the utilitarian product (respectively $t(102)=2.596$, $p=0.044$; $t(78)=2.788$, $p=0.007$). There are no differences between the two groups (experimental and control) with regard to the perception of the hedonic/utilitarian product as being hedonic except for the 2nd study (utilitarian good experiment) in which the use of a chatbot seems to

Table 2 Hedonic and utilitarian perception before-after stimuli in the hedonic and utilitarian product

Variable*	Study 1 experiment with hedonic product						Study 2 experiment with utilitarian product					
	Experimental group			Control group			Experimental group			Control group		
	N	Average	SD	N	Average	SD	N	Average	SD	N	Average	SD
Utilitarian value perception (<i>before</i>)	52	3.58	.69	52	3.46	.54	40	3.6	.83	40	3.5	.88
Utilitarian value perception (<i>after</i>)	52	4	.61	52	3.69	.67	40	4.2	.77	40	3.6	.98
Hedonic value perception (<i>before</i>)	52	3.4	.69	52	3.3	.59	40	3.4	.75	40	3.2	.96
Hedonic value perception (<i>after</i>)	52	3.7	.66	52	3.6	.63	40	3.8	.81	40	3.5	.88
Customer purchase intention	52	3	.92	52	2.5	1	40	3.5	1.03	40	2.9	1.16

p* value < 0.1, *p* value < 0.05, ****p* value < 0.01

* (judgement in a 5 point Likert-scale form 1 = completely disagree; 5 = completely agree)

increase the perception of hedonicity only for utilitarian products although with a marginal significance ($t(78) = 1.719$, $p = 0.090$). Therefore, regarding the role of the chatbot in modifying the perception of hedonic products, it emerges that, on average, following the use of the chatbot the experimental group considers the hedonic products much more functional compared to the group that consulted only the website (with a significant increase in the difference in the perception of functionality of cruise products compared to the +8.4% control group). H1a is supported. This indeed confirms the results of the literature that states how the internet strengthens the perception of functionality of products during online shopping (To et al., 2007). Indeed, the possibility to ask for information, to improve one's knowledge of the services offered by the brand makes hedonic products online more practical, functional and encourages purchase intention.

This consideration is coherent with the recent literature on the topic (Liao et al., 2016). Greater functionality is also perceived in the interaction with the conversational agent of the telephone service brand (with a significant increase in the difference of the perception of functionality of telephone service products compared to the control group of 16.7%). H1b is therefore supported. Finally, as you can see from Table 2, the chatbot, albeit with only slight significance, increases the perception of "hedonicity" but only for utilitarian products of +7.9% compared to the control group. Therefore, the presence of a chatbot could be the tool capable of "entertaining" the potential customer, proving it to be a valid auxiliary channel at the service of the relationship with the customer. Therefore, H1d is supported while H1c is not supported.

Table 3 illustrating the differences in the experimental group and in the control group before and after the stimulus, confirms that the chatbot in both hedonic and utilitarian products, manages to shift the perception of value towards a more hedonic component (respectively $t(102) = 2.462$ $p = 0.015$, $t(78) = -2.291$ $p = 0.025$) and more functional (respectively $t(102) = 3.447$ $p = 0.001$; $t(78) = -2.787$ $p = 0.007$), proving it to be a valid instrument at the service of marketing strategies.

Table 3 also shows that, only for the hedonic product, even the control group, after just browsing the website, attributed a greater value of hedonicity to the product cruise products ($t(102) = -2.081$ $p = 0.040$). This is most likely due to the fact that the website already has a good component of hedonicity and this may have encouraged groups towards a more hedonic perception during navigation. Therefore, except for this last point, Table 3 confirms that it is only in the experimental group that there is a change in the perception of the hedonic and utilitarian product towards a greater degree of solidity and/or functionality. Contrary to recent literature it was found that "there are not many hedonic values that could influence Internet shopping and one of the major reasons is that Internet shopping is unable to provide the diversified sensational simulation that physical shopping does" (To et al., 2007). We demonstrated that chatbot is an important tool that increases the perceived hedonic value of the product by substituting in part, the physical vendor. Consequently, a firm that wants to improve the experience of the customer while he browses through the website should invest in the chatbot to boost the effectiveness of the website.

Table 3 Difference between before and after the experiment on the value perception of hedonic and utilitarian products

	Before			After		
	<i>N</i>	Mean	SD	Mean	SD	Sig
<i>Study 1: experiment on hedonic products</i>						
<i>Experimental group</i>						
Hedonic value perception	52	3.4	.69	3.7	.66	.015**
Utilitarian value perception	52	3.5	.69	4	.61	.001***
<i>Control group</i>						
Hedonic value perception	52	3.3	.59	3.6	.08	.040**
Utilitarian value perception	52	3.4	.54	3.7	.09	.057
<i>Study 2: experiment on utilitarian products</i>						
<i>Experimental group</i>						
Hedonic value perception	40	3.4	.74	3.8	.81	.025**
Utilitarian value perception	40	3.6	.83	4.2	.77	.007***
<i>Control group</i>						
Hedonic value perception	40	3.2	.95	3.5	.88	.118
Utilitarian value perception	40	3.5	.87	3.6	.98	.632

** p value < 0.05, *** p value < 0.01

4.2 Chatbot and customer purchase intention

As for the role played by the chatbot during the purchase process of hedonic and utilitarian products, interesting results emerge. With hedonic and utilitarian products a significant difference is found in the averages ($p = < 0.05$) in the behaviour of the experimental group (respectively $M = 3$, $SD = 0.92$; $M = 3.5$ $SD = 1.03$) with respect to the control group (respectively $M = 2.5$, $SD = 1$; $M = 2.9$ $SD = 1.16$). The chatbot stimulates the intention to purchase in the experimental group both as regards the purchase of the hedonic product and the utilitarian one (respectively $t(102) = -2.040$ $p = 0.044$; $t(78) = -2.138$ $p = 0.037$) (see Table 2). Figure 1 shows the main results. In fact, to the question: "How likely is it that you will buy what you found today on the internet in the next 12 months", the group that used the chatbot showed greater interest in the hedonic product it was consulting than those who only gathered information by browsing on the website. Those who used the chatbot demonstrated a higher propensity to purchase than those who limited themselves to browsing the website (with an increase in purchase intent, compared to the control group, of 20%). The H2a hypothesis is supported.

As for the role of the chatbot in the intentions to purchase utilitarian products, even in this case, on average, the experimental group responded much more positively ($p = < 0.05$) to the possibility of purchasing the product they saw and requested chatbot information with respect to the control group. In fact, the experimental group declares that it will probably buy the mobile phone product/service seen on the telephone service brand website next year. To the question: "How likely is it that you will buy the product/mobile phone service seen on the telephone service

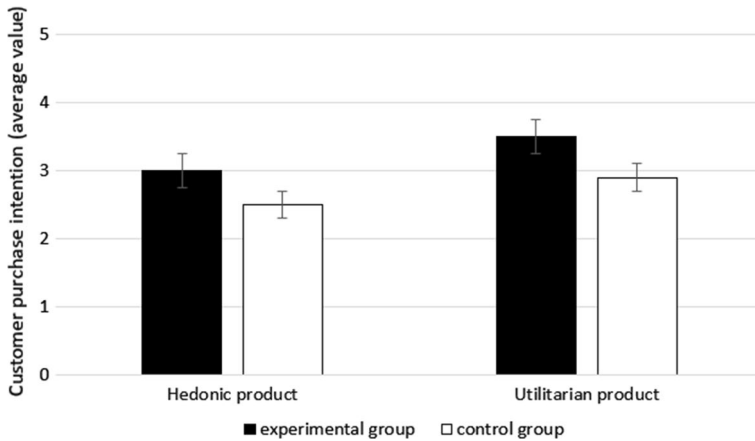


Fig. 1 Level of intention to buy per condition

brand website next year" the group that used the chatbot showed greater interest in the product they were consulting than the group that collected information just by browsing the telephone service brand website. The interaction with the chatbot has a positive impact on the intention to purchase the utilitarian products just visited on the website (with an increase in the propensity to purchase, compared to the control group, of 20.7%). The H2b hypothesis is supported.

4.3 The importance of familiarity and previous experience with chatbot on customer purchase intention

Finally, we tried to understand the importance of familiarity with the brand and previous experience with the use of a chatbot on the intention to purchase the cruise and telephone service products. Table 4 therefore shows the significance of the overall differences (i.e. on the total of respondents) and compared to the experimental group and the control group. As you can see from Table 4, a previous experience with a chatbot has no effect on purchase intent. Instead, on the whole, familiarity is still an important variable for the consumer's intention to buy the telephone service product, while it is not important in the hedonic brand product (respectively $t(102) = -0.773$ $p = 0.441$; $t(78) = -2.441$ (0.017)). If we compare the results between the experimental and the control group, familiarity is not decisive in the experimental group while it is important for the purchase in the control group both for purchase of the hedonic product (even if minimal since $p < 0.1$) and the utilitarian one ($p < 0.05$). We can therefore confirm the hypotheses H3b H3c according to which the use of a chatbot compensates for low familiarity with the brand, increasing the propensity to purchase, especially in those users who know less about the brand, while familiarity with the brand is an important factor for those who have only browsed the website (control group), for whom familiarity is a determining factor for the propensity to purchase. In the group that had an interaction with the chatbot, therefore, the factor "familiarity with the brand" does not seem to be decisive. This is verified both with

Table 4 Role of familiarity and previous experience on customer purchase intention in the hedonic and utilitarian product

Variable	Experimental + control groups				Experimental group				Control group			
	N	Average	SD	t (Sign)	N	Average	SD	t (Sign)	N	Average	SD	t (Sign)
Study 1: experiment with hedonic product												
Customer purchase intention												
High familiarity	32	3.01	1.10	-.773 (.441)	17	2.9	.95	.831 (.410)	15	3.46	.915	-1.710 (.090)*
Low familiarity	72	3.18	0.93		35	3.17	.89		37	2.86	1.22	
Chatbot experience	55	3.01	1.07	.502 (.617)	29	2.96	.905	1.135 (.262)	29	2.96	1.20	.501 (.619)
No-chatbot experience	49	3.12	1.03		23	3.26	.963		23	3.13	1.14	
Study 2: experiment with utilitarian product												
Customer purchase intention												
High familiarity	63	3.77	.85	-2.441 (.017)**	35	3.71	.95	1.138 (.262)	28	3.85	1.19	-2.287 (.028)**
Low familiarity	17	3.17	1.07		5	3.20	.83		12	3.16	0.70	
Chatbot experience	59	3.66	.957	-.177 (.860)	5	3.60	.945		35	3.62	.972	.385 (.702)
No-chatbot experience	21	3.61	.864		35	4.00	1		5	3.80	.447	

*p value < 0.1, **p value < 0.05, ***p value < 0.01

the hedonic product and with the utilitarian product (respectively $t(102)=0.831$ $p=0.410$; $t(78)=1.138$ $p=0.262$).

Familiarity in those who used the chatbot is therefore not decisive. H3b and H3c hypotheses are supported. This has important managerial implications: the chatbot can increase the propensity to buy even when the user does not have strong familiarity with the brand thanks to the acquisition of functional information on the product. For those who have not used the chatbot, there are differences in the intention to purchase among those who have low or high familiarity. In fact, those with a high level of familiarity have a greater intention to buy the product than those with low familiarity. This confirms the reference literature that affirms the importance of familiarity in intention to purchase (e.g. Laroche et al., 1996), but at the same time advances the research on the subject since it shows that the presence of an auxiliary channel during the purchasing process can contribute to the perception of the brand and reduce the importance of the brand when browsing the website. Table 5 summarizes the hypotheses and reports the test results.

5 General discussion

This research shows that the chatbot is a valuable tool during the online shopping that supports internal channels such as official websites. Moreover, it changes the overall value perception of products towards a utilitarian value thus confirming the academic literature on the topic. The utilitarian product is enhanced thanks to interaction with the chatbot which influences customer purchase intention. This paper demonstrates that chatbots for both hedonic and utilitarian products reduce the need for brand awareness. This is due to the fact that instant interaction with the brand is possible and in an informal and direct way thus reducing the “barriers” between the customer and the brand.

Today the web offers space for the creation and sharing of knowledge which is easily accessible through the new communication channels such as smartphones and tablets, paving the way for new forms of interaction between customers and organizations. The results of this study show that the conversation mediated by conversational agents are valid tools not only for relationship marketing but also for transactional marketing. In fact, the use of conversation, applied to the sales process, can develop new digital relational sales models that facilitate, shorten and finalize the transaction by moving it from the purely individual plan to a “shared” level and therefore more subject to the psycho-socio behavioral dynamics that invest the cognitive, emotional and social dimension of customers. Experiments have shown that interaction through chatbots can contribute to the purchase intention process. Furthermore, the research highlights the ability of chatbots to stimulate the buying process regardless of familiarity with the brand, which has not yet been fully explored in the conversational marketing literature. In particular, our results confirm the literature findings about the neutral role of product familiarity during chatbot conversation (Liebrecht et al. (2020). Indeed, in our results, the low/high familiarity with the brand is significantly relevant to purchase intention for the control group (which only browses the website), rather than for the experimental group which instead

Table 5 Hypotheses testing

H	Hypotheses	Sign	Result
H1 _a	The chatbot increases the utilitarian value of hedonic products	.011***	Supported
H1 _b	The chatbot increases the utilitarian value of utilitarian products	.007***	Supported
H1 _c	The chatbot increases the hedonic value of hedonic products	.292	Not supported
H1 _d	The chatbot increases the hedonic value of utilitarian products	.090*	Supported
H2 _a	The use of a chatbot while browsing a website has a positive influence on the propensity to purchase in hedonic products	.044**	Supported
H2 _b	The use of a chatbot while browsing a website has a positive influence on the propensity to purchase in utilitarian products	.037**	Supported
H3 _a	The previous experience with a chatbot has a positive influence on the intention to purchase	.262	Not supported
H3 _b	familiarity with a hedonic product does not change the purchase intention during the consumer-chatbot conversation	.410	Supported
H3 _c	familiarity with a utilitarian product does not change the purchase intention during the consumer-chatbot conversation	.262	Supported

*** < .001, ** < .01, * < .05

interacts through the chatbot. This would suggest that chatbots may have a mitigating role on brand awareness and they could be adopted as tools to activate lead strategies. In view of the potential of these types of communication it is therefore important to invest resources and knowledge to verify the degree of utility with respect to the type of business and the type of target. Therefore, the challenge is to invest in the development of chatbots based on artificial intelligence and on a strong integration with company information systems. This work highlights an opportunity to further the research on a topic of current interest for both academics and practitioners. Two directions for research seem particularly promising. Firstly, given that the present study has considered the role of chatbots on the intentions to purchase hedonic and utilitarian products, extending the analysis to live chats is an aspect to be explored. Secondly, it would be useful to investigate the impact of chatbots on the intention to purchase at all stages of the customer journey. In fact, the chatbots under investigation support the customer when he enters the website but there is nothing to stop the customer from interacting with the chatbot during his online purchase making the purchase process even more experiential (Hoyer et al., 2020). It could therefore be useful from the managerial point of view to extend the analysis by identifying the effect of chatbots during the initial phases of contact with the potential customer and then, in the more specific purchase phases.

5.1 Theoretical implications

This paper adds to the growing body of knowledge in service marketing about the use of digital services, especially chatbots, in service interactions. We show that chatbots have a role in the perception of the hedonic and utilitarian product value and can be a relevant factor for the customer's decision-making process. From a theoretical perspective, the present study contributes to the service marketing literature in three ways. Firstly, this research is, to the best of our knowledge, the first study that empirically studies the role of chatbots on the perception of the value of a hedonic and utilitarian service when used as a service contact tool by the organization in the context of conversational commerce. In doing so, the study enriches the as yet limited research by investigating the role and importance of contacting the customer through chatbots during the online customer experience from the consumer behaviour perspective (Sampson & Chase, 2020). These results are also consistent with what Hoyer et al. (2020) say when conceptualizing the importance of new technology during the customer journey: they illustrate how chatbots are important in the initial phase of the customer journey (pre-transaction). This suggests that managers should adopt chatbots as support services since they are the kind of technology that points to a cognitive dimension. Nonetheless, we have demonstrated that for a service of a functional nature, chatbots can make perception more hedonic and as such make the customer journey more experiential. This is consistent with what has been stated, but not demonstrated by To et al. (2007).

Secondly, by measuring the degree of familiarity with a brand before and after the use of the chatbot we contribute to recent research which supports the importance of knowledge and familiarity with digital services. In particular, we are consistent

with Danckwerts et al. (2019) who provided initial evidence that the conversational agent, in digital services of a hedonic nature, can elicit a sense of belonging thanks to the perception of humanity and personalization that feeds proactive behaviours by the customer towards the service, such as that of buying a premium service. Our results also show that chatbots are able to lower the need for familiarity with the brand thanks to their ability to arouse credibility. This study therefore extends the studies investigating the role of chatbots and virtual agents "as product advisors in e-commerce" (Tan & Liew, 2020).

Thirdly, we provide evidence of the use of chatbots on the intention to purchase in hedonic and utilitarian contexts by adopting a field experiment that allows you to see the real consumer experience for yourself. We believe that our research contributes to the studies that investigate the role of e-service agents for the online customer shopping experience and therefore also the role of conversational commerce through service robots. Additionally, the proposed experiment and its results can be useful in omni-channel strategic planning for future customer relationship marketing strategy research. This research may also be used as a basis to determine how hedonic and utilitarian perceived value may be stimulated by chatbot conversations. Digital assistants may change the online shopping context increasing the hedonic value experience but at the same time reducing the time for decision making by reinforcing the utilitarian value perception of the utilitarian and hedonic products. Chatbots during the online shopping saves the customer's time, minimizes effort expectancy, is easy to use and minimizes personal hassle in shopping. This study contributes to the literature on brand relationship by affirming that with conversational commerce, the relationship is not the final stage of the purchasing process but becomes the means by which to proceed to the purchase. This study is coherent with findings in the literature (Fournier, 1998) according to which the brand-customer relationship based on commitment is the most effective way to proceed with the purchase. In line with this stream of research, the present study contributes to advance scholarly knowledge by showing that the conversations can be an effective contact channel to activate positive online customer experiences.

5.2 Managerial implications

This research could support digital marketing manager in their effort to improve the effectiveness of official websites leading the customer towards a greater propensity to purchase. Companies should design and carefully manage chatbots by monitoring user involvement and quality of experience in order to feed a calibrated continuous improvement process on company objectives: attention to time, attention to the tone of the communication, attention to the quality of the information provided, attention to the training of contact personnel, control of the channels of instant messaging applications by feeding ongoing relationships with their customers that could have an effect on retention, the use of conversational marketing even in the final stages of the customer journey so as to activate shortcuts along the channel thanks to the engagement lever. Furthermore, since the chatbot mitigates familiarity with the brand, companies should equip themselves with chatbots that are able to return

useful information to the customer. The results of this research encourage practitioners to adopt chatbots to engage in a conversation with the customer accessing the website since their presence increases the value of the brand and particularly, in utilitarian products, increases the hedonic value of the product. In fact, companies could improve the consumer's online experience by implementing conversations with intelligent chatbots since they will soon be able to elicit feelings of fun and pleasure by facilitating consumers' propensity to purchase. Furthermore, companies could benefit from the use of chatbots which, performing a function similar to that of the physical seller, can easily direct the customer towards purchases aimed at his needs. Therefore, there is a need for companies to acquire adequate digital and artificial intelligence skills and knowledge in order to make the most of the information coming from interactions with chatbots and formulate digital relational sales strategies more accurately.

5.3 Limitations and future research

This study is obviously not free of limitations. First, the experiments were conducted on a sample of young university students without considering potential differences deriving from the participation of consumers belonging to other age groups. The second limitation is connected to the choice of cruise sector which may not be fully consistent with the target used for the study. Consequently, this could have influenced the assessment of the purchase intention of the subjects involved in the experiment. Moreover, the research was conducted in a single country, Italy, without taking into account that the nationality and culture of the participants could generate differences in consumer behavior. Finally, the present study mainly relies on self-stated customer perceptions; future studies might be based on company data from real interactions with customers. As for future research developments, it is believed that further research will improve the quality of the work carried out so far. For example, it could be useful to increase the sample size, to involve more types of customer segments, to identify metrics to detect the effectiveness of conversational commerce on customer loyalty and to detect the impact of the variables that affect communication with the customer (tone, language, graphics, etc.). Given the growing development of chatbots on the websites, other research questions still need an adequate response: What role does content play in interaction and engagement? What is the role and importance of conversation on customer loyalty? What is the role of satisfaction in interacting with a chatbot by word-of-mouth? Although the sample was collected randomly, the sample was composed mainly of young customers and this limits the generalizability that a broader population can give. This research refers to two specific sectors, for this reason, a further research should extend the investigation individuating the different transversal sectors. The findings of the study are based on laboratory experiments and not on actual purchase situations, although a scenario was given. In future studies using field experiments could be carried out in order to overcome this limitation.

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